



UNIVERSITY OF SCIENCE
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Writing Paper's Results and Discussion Sections

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Outline

- Results
- Discussion
- Conclusion
- Exercise

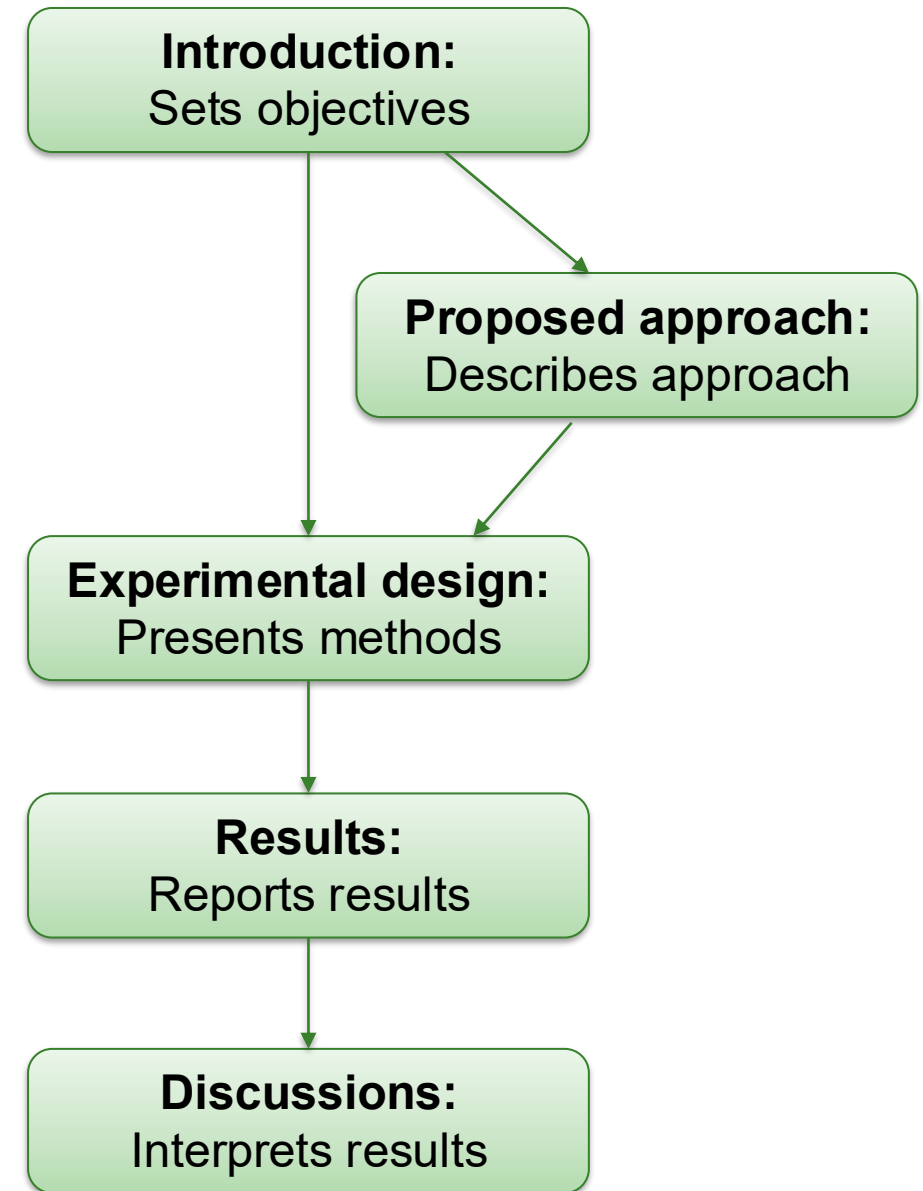
Organizing (conference paper)

- Title (1000 readers)
- Abstract (4 sentences, 100 readers)
- Introduction (1 page, 100 readers)
- The problem (1 page, 10 readers)
- My idea (2 pages, 10 readers)
 - Proposed approach
- The details (5 pages, 3 readers)
 - Experimental design
 - Results
 - Discussions
 - Threats to validity
- Related work (1-2 pages, 10 readers)
- Conclusions and further work (0.5 pages)

Source: “[How to write a great research paper - Seven simple suggestions](#)” by Simon Peyton Jones, Microsoft Research, Cambridge

Results

- Objective
 - Reporting the results obtained
- Authors need to answer the question
 - What did they find?
- Results are intended to answer research questions, hypotheses presented in the Method section.
 - Results must reflect the purpose of the study given in the Introduction section



Results

- Focusing on reporting and describing information, results obtained from the experiment and research methodology
- Avoid explaining or interpreting the results or findings. Leave this for the discussion section
 - Just present facts
 - Avoid using “suggest”, “indicate”, “reveal” words when describing results in this section
 - Do not interpret whether the number is large or small, bad or good
 - Doing this is difficult sometimes
- In short papers, you can combine the results and discussion sections into one section

Results

- Organize important results into subsections (e.g., order of importance)
- Use charts, tables, diagrams, photographs, etc. to present results
 - Careful in choosing what illustrations to present results
 - Unimportant results should be left out of charts, tables, diagrams. You can describe them using texts
 - Charts, tables, etc. must have captions and must be referenced in your text.
 - Do not leave any illustrations without being referenced in the paper
 - Cite the source in the caption if a figure, diagram, etc. is copied from a source
- In your text, do not just simply repeat or list what are presented in tables, figures

Results

- Do not leave tables, figures, etc. speak for themselves. Describe them in your text!
- In your text, do not just simply repeat or list what are presented in tables, figures, etc.
 - Provide additional information besides data shown
 - Describe complex tables (e.g., what each column means)
 - Describe noticeable/important results, and results linked to the objectives
- Use past tense to describe what you did
 - E.g., “The experiment was carried out on mobile devices”
 - It is ok to use present tense sometimes to describe results

Results

■ Describe a table (its columns and values)

Table 6
Global windowing calibration results for COCOMO-202.

Window size	Projects estimated	SA (window) (%)	SA (no window) (%)	Difference in SA (%)	MAR (window)	MAR (no window)	Difference in MAR	Percent Difference in MAR (%)	p-value (MAR)	Cohen's d
5	122	66.1	63.8	2.3	180.4	192.9	-12.5	-6.5	0.0001	0.44
10	249	63.3	62.9	0.4	195.7	197.9	-2.2	-1.1	0.0018	0.24
15	266	67.9	63.5	4.4	171.1	194.6	-23.5	-12.1	0.0007	0.15
20	258	72.4	68.0	4.4	147.3	170.6	-23.3	-13.6	0.0000	0.27
25	231	78.2	73.6	4.6	116.2	140.7	-24.5	-17.4	0.0000	0.35
30	231	78.0	73.6	4.4	117.3	140.7	-23.4	-16.6	0.0006	0.12
40	184	79.5	75.4	4.0	109.4	130.9	-21.5	-16.4	0.0003	0.27
50	184	79.4	75.4	4.0	109.7	130.9	-21.3	-16.2	0.0001	0.34
60	160	76.4	74.7	1.7	125.7	134.6	-8.9	-6.6	0.0211	0.11
70	146	80.2	76.1	4.1	105.6	127.3	-21.7	-17.1	0.0040	0.20
80	146	76.1	76.1	0.0	127.3	127.3	0.0	0.0	0.0903	0.07
90	146	76.5	76.1	0.4	125.0	127.3	-2.3	-1.8	0.0602	0.15
100	146	77.2	76.1	1.1	121.7	127.3	-5.6	-4.4	0.0863	0.13
110	146	74.5	76.1	-1.6	136.0	127.3	8.7	6.8	0.4243	0.05
120	137	75.7	77.1	-1.5	129.7	121.9	7.8	6.4	0.2434	0.03
130	137	75.5	77.1	-1.6	130.4	121.9	8.5	7.0	0.5432	0.07
140	102	76.5	79.2	-2.7	125.3	111.0	14.3	12.9	0.0747	0.16
150	83	72.7	76.6	-3.8	145.2	124.9	20.3	16.3	0.0404	0.21
160	83	71.2	76.6	-5.4	153.7	124.9	28.8	23.1	0.0266	0.16
170	83	71.9	76.6	-4.7	149.8	124.9	24.9	20.0	0.0217	0.21
175	83	69.8	76.6	-6.8	161.1	124.9	36.2	29.0	0.0007	0.44
180	83	74.3	76.6	-2.2	136.7	124.9	11.8	9.5	0.7284	0.13
190	67	77.7	77.9	-0.2	119.0	117.8	1.2	1.0	0.0155	0.22
200	67	79.0	77.9	1.1	112.0	117.8	-5.9	-5.0	0.5466	0.28
210	58	86.1	85.5	0.6	74.3	77.4	-3.2	-4.1	0.4388	0.16
220	58	85.8	85.5	0.3	75.6	77.4	-1.8	-2.4	0.1477	0.34
230	46	84.9	85.1	-0.1	80.3	79.5	0.8	1.0	0.3564	0.29
240	26	85.3	84.8	0.4	78.5	80.8	-2.3	-2.9	0.1574	0.21
250	16	89.3	89.4	-0.1	57.1	56.5	0.6	1.1	0.7436	0.17

The first column in Table 6 shows the window size or the number of projects used for training with each window. The second column shows the number of projects that could be evaluated by the model calibrated using the respective window size. When a window is large, which means more projects being used for training, the number of projects evaluated is less. The third to fifth columns show the SA values for the estimates with and without a window, respectively, and their percent differences for the same number of projects specified in the second column. Negative percent differences in SA indicate that SA with a window being used is less than that without a window (positive numbers indicate the opposite). As the SA values are much closer to 100% than to 0%, the calibrated models (with and without a window) significantly out-perform random guessing. When the window size is less than 100 projects, SA with a window is slightly greater (better), improving up to 4.6%, than that without a window, but it is smaller (worse) when the window is larger (negative percentages).

Vu Nguyen, Barry Boehm, and LiGuo Huang. "Determining Relevant Training Data for Effort Estimation Using Window-based COCOMO Calibration." Journal of Systems and Software, 147, 124-146. 2018

Results

- Report statistics in parentheses ()
 - E.g., p-value, standard deviation

“As shown in Table 6 and Fig. 4 (a), global calibration using windows of 78 projects or less improves AR over retaining all projects completed so far, and the improvements are statistically significant (p -value < 0.05) with the percent difference in AR averaging 12.8%. At most windows in this range, the Cohen's d values indicate that the differences are real but with small effect sizes (Cohen's $d \geq 0.2$). The best improvement in AR is achieved with the window size of 32 projects, reducing MAR by 18.2% compared to using no window. Larger windows of between 150 and 190 projects produce worse MAR, and at most windows, the differences are statistically significant (p -value < 0.05) with the average percent difference reaching 19.1%. Out of these ranges, window sizes of between 79 and 149 and more than 190 projects, the differences in AR are not statistically significant.”

Outline

- Results
- Discussion
- Conclusion
- Exercise

Discussion

- Purpose: interpret, explain, and compare your results with other studies
- The results section describes the facts while the discussion section interpret the results
 - You can present your opinions in this section
- Typical structure
 - Summarize hypotheses, research questions, objectives, and findings in the first paragraphs
 - Compare your results with those of previous studies
 - Generalize and present the implications of results. Provide explanations
 - Discuss advantages and limitations of the study

Discussion

- Limitations and issues your approach/implementation currently cannot address
 - Optional: how are you going to address them in future work
- Other caveats (scope of your approach)
- It is often a good idea to list (obvious) limitations and discuss possible solutions for them rather than hiding them
 - Reviewers can often identify obvious limitations even if you don't state them; then they will criticize your work on these limitations (you often don't have a rebuttal against these criticisms in conference reviews).
 - If your paper discusses these obvious limitations as well as their potential solutions, the situation can be alleviated (it is like you have a rebuttal in your paper already before being criticized!)
- Possible applications of your approach that you haven't validated but are convincingly feasible or effective.

Discussion

- In writing a discussion section, keep the following points in mind
 - ❑ Identify significant patterns in the data and relationships between variables. Offer tentative explanations for these patterns and relationships.
 - ❑ Compare the actual data produced with any predictions or questions posed in the introduction section
 - ❑ If any of the results differ from the expected results, offer possible explanations for the discrepancies. Present the most probable explanations first and the least probable last.
 - ❑ Consider how well the data answer any research questions.
 - Do the results answer the questions completely? If not, explain what questions still need to be addressed and give possible explanations why the results may be inconclusive.

The Mayfield Handbook of Technical & Scientific Writing

Leslie C. Perelman, James Paradis, Edward Barrett, <https://www.mhhe.com/mayfieldpub/tsw/abstract.htm>

Discussion

- You should avoid
 - ❑ Repeating what have been described in the Results section
 - ❑ Explaining results incorrectly
 - ❑ Missing important results
 - ❑ Too-short or too-long explanations
 - ❑ Focusing on only one reason and miss other possible reasons for results
 - ❑ Inconsistent interpretation
 - ❑ Criticizing previous studies while your results do not support the points you criticize

Discussion

- You can cite the research questions and provide answers

5. Discussion

In this section, we provide our discussions and interpretations of the results presented in the previous section.

5.1. Answers to the research questions

The research questions in this study were focused on investigating the effect of windows on estimation accuracy of COCOMO II calibration. They were addressed by running the calibration process and comparatively assessing the calibrated models using three data sets, COCOMO-292 for global calibration, COCOMO-164, and NASA-93 for local calibration.

RQ1: Is there a difference in estimation accuracy between the COCOMO model calibrated using all available completed projects and the COCOMO model calibrated using a window of only n most recently completed projects?

Research question RQ1 was answered by testing the null hypothesis which assumes that there is no statistically significant difference in estimation accuracy between the model calibrated using the training set of all projects completed so far and the one calibrated using the window of n most recently completed projects. This research question was addressed from both global and local calibration perspectives. The results from global calibration (see

Conclusion

- Often easy to write conclusions
 - Nothing here should surprise readers; simply summarize your contributions and findings
 - In the introduction, “We propose a new approach ...” vs. In the conclusions, “We have proposed a new approach ...”
- You can state the broader impacts of your approach
 - Implications
- You can optionally describe limitations and future work here if you don’t have a discussion section for them and propose future work

Slide adapted from Tao Xie

Conclusion

We have presented a study that suggests that maintenance work consists of three fundamental activities focused on forming, navigating, and manipulating a working set of task-relevant code fragments. Our findings are largely consistent with previous studies of program comprehension: programmers' understanding is facilitated by recognizable patterns in code [3, 6]; expert programmers often start understanding code top-down, but finish bottom-up [5]; and programmers follow an as-needed strategy for navigating dependencies [9]. Our study augments these findings with a higher-level account of maintenance work, and specific data on the impact of the Eclipse IDE on programmers' maintenance task performance. In particular, our study found that on average, programmers spent about 35% of their time with the mechanics of navigating between dependencies.

Of course, our study also suffers from several limitations. The 10 programmers in our study are not likely to be representative of programmers in industry. The size of the Paint program is certainly not representative of heavily maintained software systems. Programmers that work on teams may have different strategies for maintaining code that do not involve forming working sets of task-relevant code; for example, they may be given a set of code to maintain and have to collaborate with other maintainers if their tasks take them outside of this set. Further studies of maintenance work are required to verify the generalizability of our findings.

Implications and future work

Despite such limitations, our study's findings have directly inspired several novel ideas for maintenance-oriented IDEs that deserve further elaboration and development. We are currently building an environment like the one portrayed in Figure 10 to support Java programs that we hope will both eliminate the interactive overhead identified by our study, as well as come closer to solving some of the fundamental difficulties of maintenance activity that we observed. We intend to evaluate its utility both in the lab and in the large to see how well the findings in our study generalize in practice. We hope that others will find the results presented in this paper useful for similar inspirations.

Summarize the study and findings

Limitations

“Eliciting Design Requirements for Maintenance-Oriented IDEs: A Detailed Study of Corrective and Perfective Maintenance Tasks” by Andrew J. Ko, Htet Htet Aung, and Brad A. Myers

Exercise

- Read and analyze different points in a paper
 - Work with your paper partner
 - Choose one paper from the list on Moodle (one of the papers on Moodle)
 - Identify different information items of each paragraph of the Results, Discussions, and Conclusions of the chosen paper.
 - Showing results, description of the result
 - Interpreting results, showing limitations, explanation of the limitations, etc.
 - Findings, future work
 - Identify what are missing in key figures/tables, NOT described in the text
- Present your results

Exercise 2

- Work with your partner write Method, Experimental design, Results of your paper.